

RGIPT

Subject: Ai

Class: 12

Time Allowed: 90 minutes

Maximum Marks: 10

All questions are compulsory unless stated otherwise.

Name: _____

Roll Number: _____

Class: 12 Section: _____

Section A: Multiple Choice Questions

Attempt all questions. Each carries 1 mark.

1. [Easy] What is the primary purpose of an activation function in a neural network? [1 Marks]
2. [Easy] Which activation function outputs values in the range 0 to 1 and is historically used for probability outputs? [1 Marks]
3. [Easy] In the context of training, what does the loss function quantify? [1 Marks]
4. [Easy] For a regression task, which loss function is commonly used? [1 Marks]
5. [Moderate] In the linearity problem, stacking multiple linear layers is equivalent to: [1 Marks]
6. [Moderate] What does the bias term (b) allow a neuron to do? [1 Marks]
7. [Moderate] During backpropagation, the error signal is propagated in which direction? [1 Marks]
8. [Challenging] Which optimizer concept is described by “feel the slope and take a step in the direction that reduces the loss”? [1 Marks]
9. [Challenging] In the simplest house price prediction example, the implied price per square foot is \$200. Which term best describes this relationship? [1 Marks]
10. [Easy] The ReLU activation function is defined as $\max(0, x)$. What is its output when $x = -3$? [1 Marks]

End of Question Paper

Answer Key:

1. To introduce non-linearity so that the network can model complex functions.
2. Sigmoid

3. How far the model's predictions are from the true values.
4. Mean Squared Error (MSE)
5. A single linear layer.
6. Shift its decision boundary.
7. From the output layer backward to the input layer.
8. Gradient Descent
9. Linear relationship
10. 0